

SIEMENS SLIDING GANTRY INSTRUCTIONS FOR USE MANUAL



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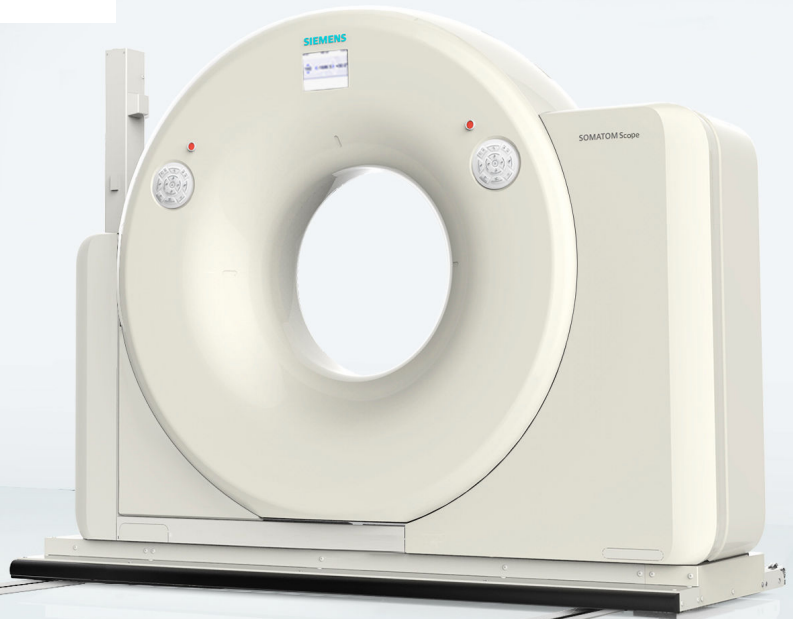
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Sliding Gantry

Instructions for Use – SOMATOM Scope

Sliding Gantry

Instructions for Use – SOMATOM Scope

Legend



Indicates a hint

Is used to provide information on how to avoid operating errors or information emphasizing important details



Indicates the solution of a problem

Is used to provide troubleshooting information or answers to frequently asked questions



Indicates a list item



Indicates a prerequisite

Is used for a condition that has to be fulfilled before starting a particular operation



Indicates a one-step operation



Indicates steps within operating sequences

Italic

Is used for references and for table or figure titles



Is used to identify a link to related information as well as previous or next steps

Bold

Is used to identify window titles, menu items, function names, buttons, and keys, for example, the Save button

Blue

Is used to emphasize **particularly** important sections of the text

Courier

Is used for on-screen output of the system including code-related elements or commands

Courier

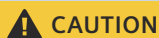
Is used to identify inputs you need to provide

Menu > Menu Item

Is used for the navigation to a certain submenu entry

<variable>

Is used to identify variables or parameters, for example, within a string



CAUTION

Used with the safety alert symbol, indicates a hazardous situation which, if not avoided, could result in minor or moderate injury or material damage.

CAUTION consists of the following elements:

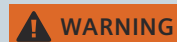
- Information about the nature of a hazardous situation
- Consequences of not avoiding a hazardous situation
- Methods of avoiding a hazardous situation

WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

WARNING consists of the following elements:

- Information about the nature of a hazardous situation
 - Consequences of not avoiding a hazardous situation
 - Methods of avoiding a hazardous situation
-



Legend

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1 Introduction

This document provides detailed information about your accessories and system components.

1.1 Compatible systems

The Sliding Gantry option is available for the following CT system:

- SOMATOM Scope (Power configuration)

1.2 Applicability

This document is valid only in conjunction with the Instructions for Use of the corresponding SOMATOM CT system.

2 Safety

For the secure operation of the accessories and system components, read the following sections.

The statutory stipulations and guidelines, as well as the information specified in this document, form the basis for ensuring the safety of people and the safe operation of the equipment.

- Read these Instructions for Use carefully and follow the instructions contained therein.
- Read the safety instructions in the CT system documentation in addition to the following safety information. For more information, see the SOMATOM Instructions for Use.

2.1 Safety information

CAUTION

Not observing the Instructions for Use of the system, system options and accessories!

Injury of the patient.

- ◆ Always use the Instructions for Use in conjunction with the Instructions for Use of the particular units used.
- ◆ Follow the safety instructions.

CAUTION

Emergency stop on CT not usable for combisystem!

Risk of injury to the patient and damage to the system.

- ◆ Note, that the emergency STOP switches at the gantry work only in conjunction with the gantry and are not related to any other equipment, e.g., the patient table. When the gantry is in the park position, all emergency stop switches are disabled.

** CAUTION**

Tripping because of uneven floor!

Risk of injury of user.

- ◆ When crossing the gantry rails, take care not to injure yourself.

** CAUTION**

Pinching of fingers in cableguide!

Risk of injury to the patient and personnel.

- ◆ Keep hands off the flexible connection lead.

** CAUTION**

Intercom only possible in CT mode!

Patient cannot be understood in emergency.

- ◆ Use the intercom system for communication with the patient only as long as the patient table is within the audible communication range of the gantry intercom.

** CAUTION**

Missing ECG-Sync due to the disconnection of the patient's ECG-electrodes!

X-ray not, or only partially, usable.

- ◆ A contrast enhanced scan should not be aborted but operated with synthetic ECG-data.

2.2 Gantry movements

CAUTION

Collision of tilted gantry and combi system!

Damage to the gantry.

- ◆ Avoid collision of the gantry and the patient, or collision of other objects in the examination room with the gantry.

CAUTION

Collision with gantry!

Uncontrolled movements of the head of the patient.

- ◆ Make sure that the head of the patient is correctly positioned.

CAUTION

Cable damage due to gantry motion!

Undesired radiation

- ◆ Remove or appropriately secure any attached optional equipment such as footswitch or ECG to avoid risk of damage or injury.

CAUTION

Collision with cable guide!

Injury to the patient or personnel.

- ◆ Avoid collision of the gantry and the patient, or collision of other objects in the examination room with the gantry.

⚠ CAUTION

Collision between gantry and bolus injector!

Pulled out catheter.

- ◆ Avoid patient collision with the gantry, or collision of other objects in the examination room with the gantry.

Collision protection

Movement of the gantry stops when the bumper at the gantry front or at the gantry back is obstructed, for example by the foot of the operator.

Patient table related advice

Because the patient table is not connected to the CT scanner, the system is unable to determine the current table position. To avoid any collision involving the patient table, particular care is necessary when operating the patient table.

For instructions using the patient table, please refer to the instructions for use of the patient table manufacturer.

⚠ CAUTION

Moving gantry instead of the table!

Risk of injury to the patient and damage to the system.

- ◆ Always be aware that the STOP keys at the CT system do not act on the table used or other devices installed in the room, only on the CT system.

⚠ CAUTION

Table top too wide!

Injury to the patient or personnel.

- ◆ Make sure that the table top is not too wide.
- ◆ Observe the technical table top specification.

 CAUTION

Moving patient table and tilting gantry!

Injury of the patient and damage to the system.

- ◆ Avoid collision areas during scanning.
- ◆ Use proper positioning accessories and secure it when raising or lowering the patient table or moving or tilting the gantry.
- ◆ Avoid collision of the gantry and the patient, or collision of other objects in the examination room with the gantry.

 CAUTION

Wrong scan position after manual table motion!

Incorrect diagnosis resulting from incorrect positioning of the tomograms.

- ◆ New Frame of Reference (i.e. new Topogram).

The collision-safe table height is determined by the specific table top width and the gantry tilt.

Observe the collision-safe table heights shown in the technical specifications.

 CAUTION

Table not in CT position!

Injury to the patient or personnel.

- ◆ Avoid collision of the gantry and the patient, or collision of other objects in the examination room with the gantry.

** CAUTION**

Sideways motion of table in gantry!

Injury to the patient or personnel.

- ◆ Move the table sideways with care.

** CAUTION**

Incorrect positioning of the patient!

Injury to the patient or personnel.

- ◆ Ensure that the patient is correctly positioned.

2.3 Measurement related safety information

** CAUTION**

Scan while table not in CT position!

Undesired radiation

- ◆ Move the table horizontally to a position in which the patient's body area of interest extends over the gantry rails region, enabling CT examinations to be performed over that body area without restriction.

** CAUTION**

Wrong entries!

Dose not as desired.

- ◆ Check scan parameters before scanning.

Positioning aids

CT positioning aids are not included in the delivery of sliding gantry systems. The accessories used with the patient table must meet CT requirements with respect to artifacts and absorption.

CAUTION

Image artifacts!

X-ray not, or only partially, usable.

- ◆ Set the table to the height and horizontal position in accordance with the technical table top specification.

Topogram/Tomogram



In order to minimize the risk of a collision, topograms should always be scanned in the direction away from the patient table.

CAUTION

Movement of the patient table after the final topogram!

Incorrect diagnosis resulting from incorrect positioning of the tomograms.

- ◆ It is the responsibility of the operator to ensure that the table is returned to its previous topogram position, both vertically and horizontally.

CAUTION

Tilted ranges in the topogram might result in shifted tomogram ranges!

Incorrect diagnosis resulting from shifted tomogram ranges.

- ◆ Confirm the current scanning position with the light indicator.

**CAUTION**

Wrong scan position after CARE Vision or manual table motion!

Incorrect diagnosis resulting from incorrect positioning of the tomograms.

- ◆ Note, that manual table movements during a CARE Vision examination are not registered by the system.

3 Description

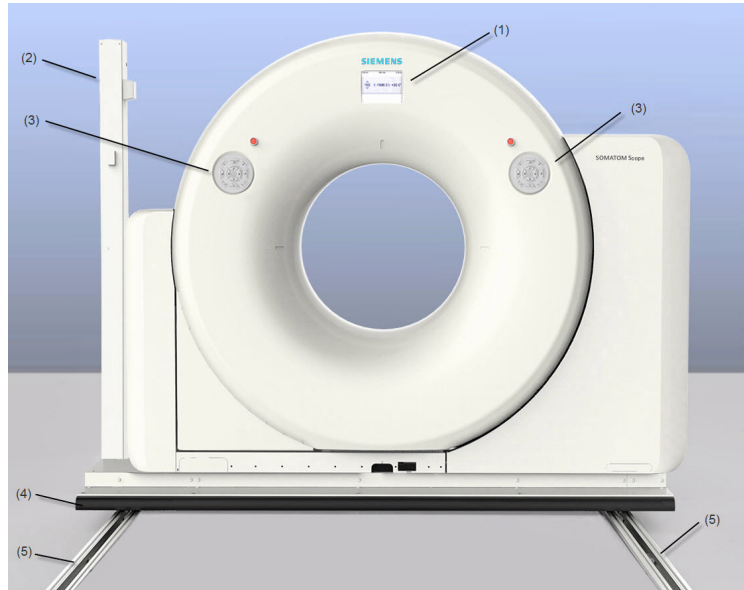
This section provides an overview of the characteristics and operating elements of the Sliding Gantry.

3.1 Purpose of the Sliding Gantry

The Sliding Gantry is available as an option and allows the operation of a CT scanner in conjunction with one or two separate patient tables. These patient tables can be part of a completely independent medical system.

3.2 System overview

The Sliding Gantry moves on rails towards, or away from, the examination table. The backmost position of the gantry can be moved to the end position in the room.



- (1) Gantry display
- (2) CT auxiliary column
- (3) Gantry operator panel
- (4) Bumper for collision protection
- (5) Rail system

3.3 Components

The following elements of the Sliding Gantry option have specific functions:

- Gantry display
- Gantry operator panels



The pictures of the components shown here are only examples. The appearance of your components depends on the system configuration.

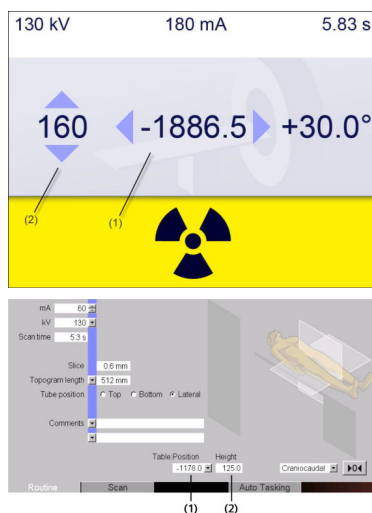
3.3.1 Gantry display

The patient table height indicator on the gantry display has no significance. The horizontal patient table position displays the gantry position in millimeters (mm) relative to the reference zero position.

The relative gantry position is also displayed on the **Routine** parameter card of your *syngo* Acquisition Workplace. Depending on the current status of the CT system, the displayed patient table height can not take every continuous value, only the predefined values are possible.

The relative gantry position is also displayed on the topogram subtask card of your software.

The patient table height indicator has no significance here as well.



- (1) Gantry position
- (2) Table height (no significance)

3.3.2 Gantry operator panels



Due to the separation of the gantry and the patient table, the table movement keys on the gantry control panel work differently.

The keys for vertical table movement are nonfunctional.

The keys for horizontal table movement are used to move the gantry along the patient table.

The **Table retraction** key is nonfunctional.



Horizontal zeroing



This key is used for horizontal zeroing. It defines the current gantry position as the horizontal reference zero position, usually an anatomic marking. The gantry position may be zeroed only during the period between the input of patient data (registration or renewal) and the start of the first scan.

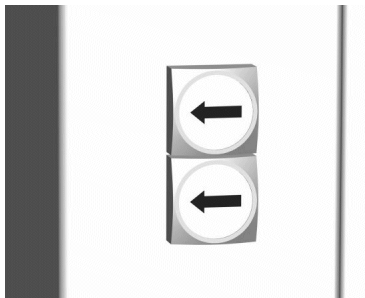
Gantry tilt



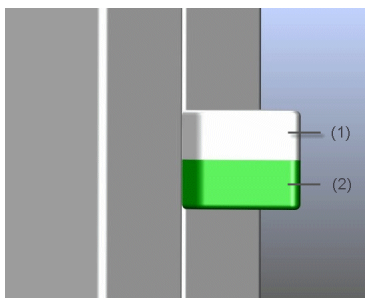
The maximum gantry tilt adjustable with these keys may be limited to less than 30 degrees, depending on your system installation.

3.3.3 Auxiliary column

The auxiliary column provides two keys for moving the gantry towards the parking position. These keys can be used in case of a malfunction of the move keys on the gantry operator panel.



Two status lamps on the auxiliary column indicate the operating state of the system:



- (1) White lamp on: gantry in parking position
- (2) Green lamp on: gantry ready for scanning and moving

3.3.4 Rail system

During scanning, the gantry slides on the dedicated rail system while the patient stays at a stationary position.

The rail system is covered by a cover strip and lateral covers (the lateral covers may not be visible if covered by flooring).



Never cross the rail system cover with a trolley if the diameter of its wheels is smaller than 50 mm.

3.4 CT mode restrictions

Because of the moving range of the Sliding Gantry and the moving speed limitations due to the stopping distance of the gantry, the CT mode is subject to the following restrictions:

- Intercom communication may be restricted as the distance between patient table and gantry increases.

3.4.1 Patient table

- Depending on the room configuration, the parameters of the patient table specified in the Instructions for Use of the CT system, may not apply to the table used with the Sliding Gantry option.
- Software functionality, such as horizontal table compensation, may be restricted.
- Table footswitch is nonfunctional.
- The **Table retraction** key on the gantry operator panel is nonfunctional.
- The separate patient table must be in a condition suitable for scanning.
- Table movement is only possible with table controls.

3.4.2 Gantry

- Depending on the room setup, gantry tilt could result in limitations for scan range.
- Table movements using the gantry operator panels are not possible.

3.4.3 CT control box

The **Unload patient** key on the CT control box is nonfunctional.

3.4.4 Measurement

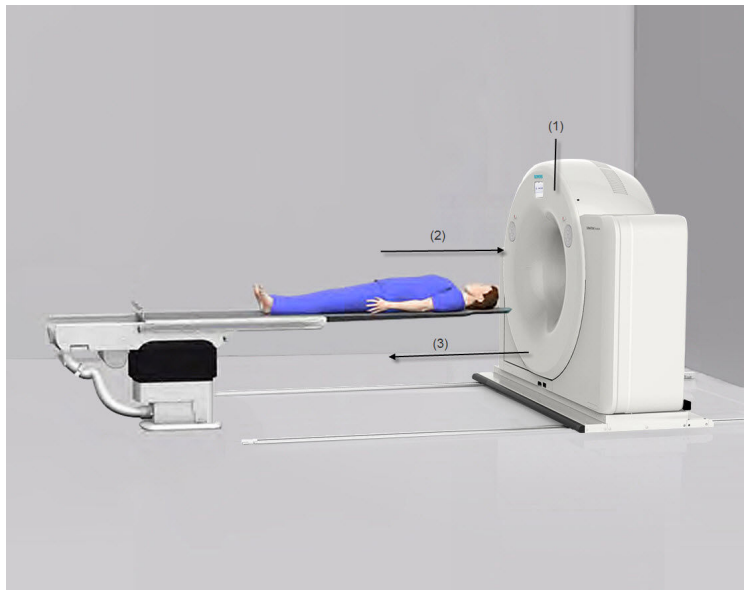
- As the osteo mattress and the osteo phantom are not available, and as it is not possible to control the table height, the *syngo* Osteo CT software cannot be used.
- Minimum cycle time during sequence scans depends on the time needed to move the gantry from one scan position to the next (cycle time may be longer than selected).
- Autorange scan is restricted to either horizontal movements or gantry tilt movements.
- Only rapid image reconstruction is possible with CARE Vision.
- Isocenter adaption is not possible as the table cannot be automatically moved vertically by the CT system.

4 Operation

This section describes the operation of the accessories and system components.

4.1 Positioning the patient

Patient orientation always refers to the gantry front side, no matter if the patient table used for examination is located in front of the gantry or behind it.



- (1) Sliding Gantry front
- (2) Sliding Gantry movement away from the patient table
- (3) Sliding Gantry movement towards the patient table

✓ Stop the patient table before you position the patient.

1 Position the patient on the patient table.



Refer to the gantry front when positioning and registering patients.

- 2 Adjust the patient table to a height which prevents any collision with the gantry.



Refer to the table Collision-safe patient table heights in the section of Technical specifications.

- 3 Move the patient table to a horizontal position in which the patient's body area of interest extends over the gantry rails region.



Only use positioning aids meeting CT requirements with respect to artifacts and absorption.

4.2 Operating the Sliding Gantry

This section provides information about operating the Sliding Gantry.

CAUTION

Moving patient table and tilting gantry!

Injury of the patient and damage to the system.

- ◆ Avoid collision areas during scanning.
- ◆ Use proper positioning accessories and secure it when raising or lowering the patient table or moving or tilting the gantry.
- ◆ Avoid collision of the gantry and the patient, or collision of other objects in the examination room with the gantry.

4.2.1 Moving the Sliding Gantry

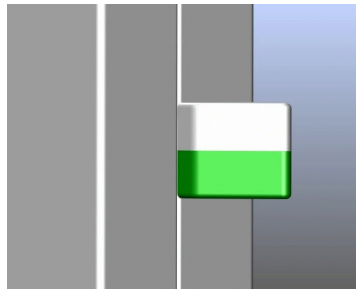
The following section describes all operations using the front position of the Sliding Gantry. The handling refers to the gantry operation and uses the front gantry operator panels.



In case of malfunction of the gantry movement keys, the rotary switch at the cable column can be used.

Moving the Sliding Gantry towards the patient table

- 1 Check the status lamps at the auxiliary column.



- 2 Press the patient table movement key to move the gantry towards the patient table. The arrow on the movement key shows away from the gantry.

When the front room end position is reached, the gantry stops automatically.

Moving the Sliding Gantry away from the patient table



- ◆ Press the patient table movement key to move the gantry away from the patient table. The arrow on the movement key shows in direction to the gantry.

When the back room end position is reached, the gantry stops automatically.

4.3 Performing the tests

This section provides information about the tests that must be performed regularly.

4.3.1 Performing the quality test

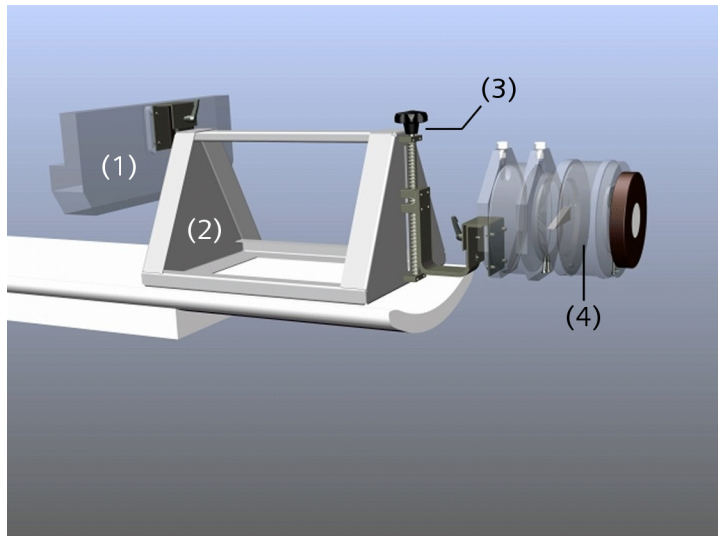
To ensure a consistently high image quality, you must perform quality measurements regularly.

A phantom set is used to perform quality measurements. You position these phantoms on the patient table prior to these tests.

The quality test must be performed with the patient table facing the gantry front.

For detailed information on performing quality measurements, see SOMATOM Instructions for Use.

Positioning the phantom set



- 1 Position the phantom holder on the patient table.
- 2 Mount the body phantom for weight balancing.
- 3 Attach the quality phantom set.
- 4 Switch on the laser light marker.
- 5 Set the appropriate patient table height.



The beam of the light marker must be aligned with the reference marking of the quality phantom set.

The quality phantom set is positioned.



Use the height adjustment of the phantom holder to adjust the quality phantom set vertically to the center of the gantry funnel.

4.3.2 Performing the system constancy test

The monthly constancy test is performed via the **Quality Constancy** dialog box of the **Siemens Med Service Software** window.

- ◆ Follow detailed instructions, see SOMATOM Instructions for Use.

4.4 Cleaning and disinfecting

Clean and disinfect the Sliding Gantry and the rail system in accordance with the cleaning instructions for corresponding accessories.

- ◆ Follow the cleaning instructions. See SOMATOM Instructions for Use.



Check regularly for the appearance of dirt and call Siemens Service if necessary.

4.4.1 Cleaning the rail system

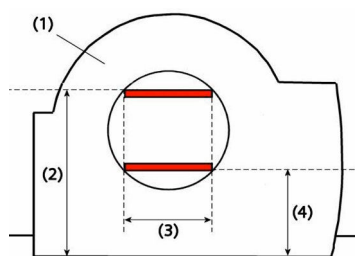
- ✓ The gantry must be stopped at a distance suitable for cleaning the cover strip.
- ◆ Clean the surface of the cover strip, the lateral covers (if visible), and the surrounding area.

5 Technical specifications

This section contains the technical specifications of the accessories and system components.

5.1 Patient table requirements

Because of the gantry funnel geometry and due to safety reasons, the minimum and maximum table height possible depends on the table top width.



- (1) Gantry
- (2) Maximum patient table height above floor
- (3) Maximum patient table width
- (4) Minimum patient table height above floor



The technical specifications of the patient table in instructions for use of your CT system do not apply to the table used with the Sliding Gantry option.

5.1.1 Patient table and table top requirements

Minimum distance of the gantry front and the patient table	The distance value depends on the patient table height and the gantry tilt. Refer to the table Collision-safe patient table heights .
CT value for the covering of the table top	< 250 HU
Attenuation of table top equivalent to aluminium at 100 kV	< 1.5 mm (excluding the connecting region of the positioning accessories)



The patient table must be possible for the patient to be moved out of the gantry in case of power loss.

Positioning accessories that are used in the region of interest should be made of special radiotranslucent material, and shaped in such a way that they cause image artifacts as minimal as possible.

5.1.2 Collision-safe patient table heights

Patient table height (mm) relative to isocenter (0°)	Maximum patient table width (mm) for gantry tilt				
	gantry tilt 0°	gantry tilt +15°	gantry tilt -15°	gantry tilt +30°	gantry tilt -30°
280	328	n.a.	n.a.	n.a.	n.a.
260	388	326	n.a.	n.a.	n.a.
240	437	390	340	n.a.	n.a.
220	477	440	420	282	n.a.
200	511	482	452	368	n.a.
180	540	517	493	431	n.a.

Patient table height (mm) relative to iso-center (0°)	Maximum patient table width (mm) for gantry tilt				
	gantry tilt 0°	gantry tilt +15°	gantry tilt -15°	gantry tilt +30°	gantry tilt -30°
160	564	546	527	482	350
140	585	571	556	523	421
120	602	592	583	556	476
100	617	609	604	584	521
80	628	623	621	606	558
60	637	633	633	623	588
40	644	641	642	635	612
20	647	646	647	644	631
0	650	648	648	646	646
-20	647	647	646	631	644
-40	644	642	641	612	635
-60	637	633	633	588	623
-80	628	621	623	558	606
-100	617	604	609	521	584
-120	602	580	592	476	556
-140	585	557	571	421	523
-160	564	527	546	350	482
-180	540	493	517	n.a.	431
-200	511	452	482	n.a.	368
-220	477	402	440	n.a.	282

Patient table height (mm) relative to iso-center (0°)	Maximum patient table width (mm) for gantry tilt				
	gantry tilt 0°	gantry tilt +15°	gantry tilt -15°	gantry tilt +30°	gantry tilt -30°
-240	437	339	390	n.a.	n.a.
-260	388	n.a.	326	n.a.	n.a.
-280	328	n.a.	n.a.	n.a.	n.a.

5.2 Sliding Gantry specifications

These values are representative for the SOMATOM Scope (power option).

Horizontal gantry travel	max. 320 cm
Examination range	depends on your installation
Automatic travel speed	up to 100 mm/s
Remote controlled feed	in min. 1.0 mm increments
Incremental positioning accuracy	repeat accuracy +/- 0.25 mm, absolute accuracy +/- 0.5 mm

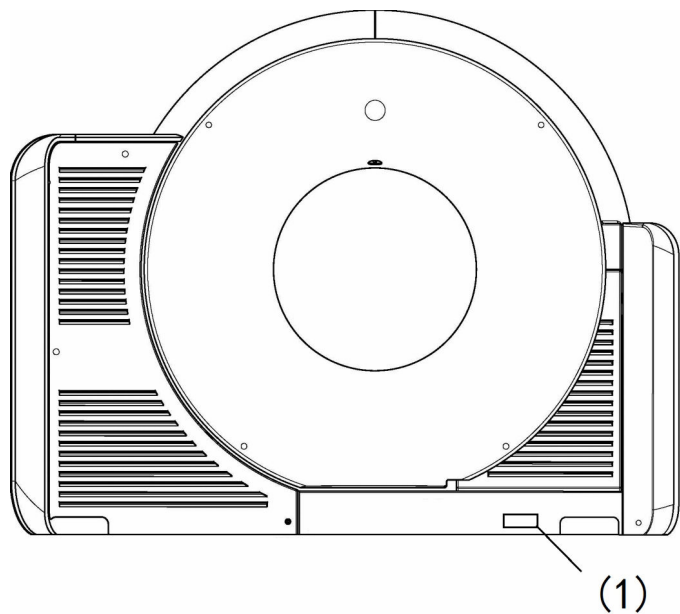
6 Location of labels

The labels can be found at the positions indicated by the reference lines and numbers in the corresponding figures.

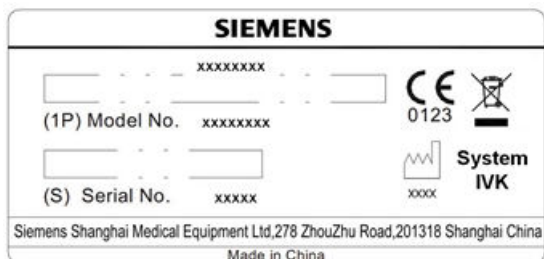


The numbers and parameters on the labels are examples only. Always refer to the original labels, which are located on the CT system and system components.

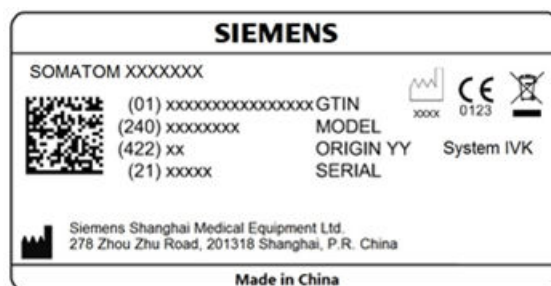
6.1 Sliding Gantry



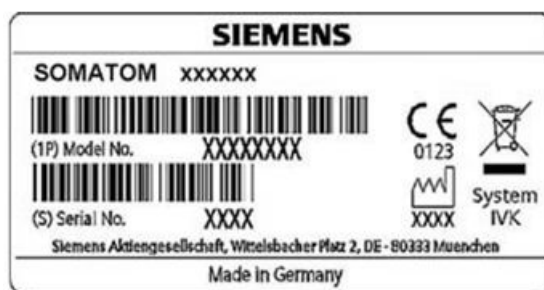
6 Location of labels



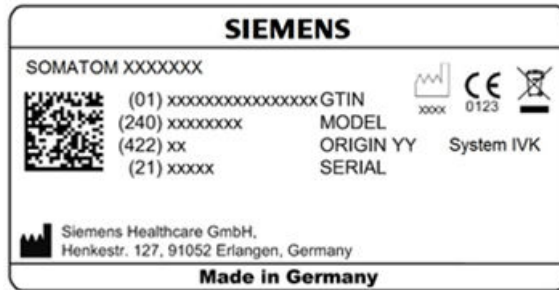
—or—



—or—



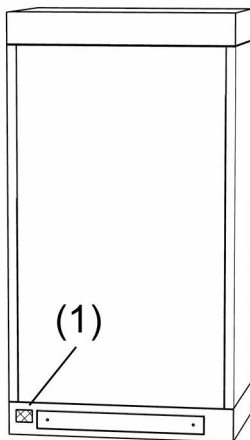
—or—



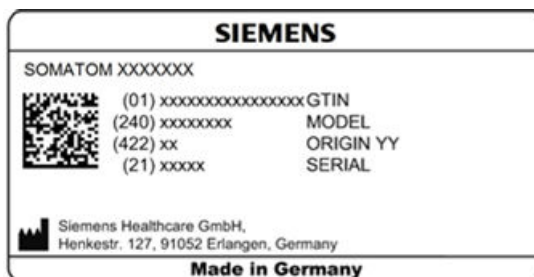
(1) Sliding Gantry label

The label is located at the rear of the gantry on the lower right side.

6.2 Control cabinet



—or—



(1) System manufacturer's identification label

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